DP Biology – D4.2

# Stability and change

- What allows ecosystems to remain stable indefinitely?
- Which human changes put ecosystem stability at risk?

SL statements D4.2.1–10. Textbook pp. 798–813. Succession is at HL.
Image: the Amazon rainforest from space. Alexander Gerst, CC BY 2.0.



# Objectives for each class

| Class | Statements | By the end of the class, you can... |
|---|---|---|
| **1** | D4.2.1–3 | Explain stability and what it requires, model it in mesocosms, and explain tipping points in the Amazon |
| **2** | D4.2.4–6 | Explain keystone species, and assess the sustainability of harvesting and farming |
| **3** | D4.2.7–10 | Explain eutrophication, BOD and biomagnification, the effects of plastics, and rewilding |

# The big picture: steady, until it tips

| Stable | At risk | Recovery |
| --- | --- | --- |
| Energy, recycling, diversity, climate | Over-harvesting, pollution, clearing | Sustainable use and rewilding |

Stable ecosystems can last thousands of years. Human pressures can push them past a **tipping point** into a new state.

**Big question**

Could the Amazon rainforest turn into savanna? What would push it there?

# Stability in natural ecosystems

D4.2.1



Tropical forest. V. Argenberg, CC BY 4.0.

**Stability** is the ability to continue over long periods. Some forests have lasted thousands, even millions, of years (Daintree, Australia).

Stable does not mean unchanging: populations fluctuate within limits.

### Local example

What evidence shows a Swiss forest has been continuous for a long time? (Tree age, dead wood, fossil pollen.)

# What stability requires

D4.2.2

| Requirement | Why |
| --- | --- |
| **Energy supply** | Usually sunlight, replacing heat lost |
| **Nutrient recycling** | Elements return to producers |
| **Genetic diversity** | Some survive new pressures |
| **Climate within tolerance** | Temperature and rain in limits |

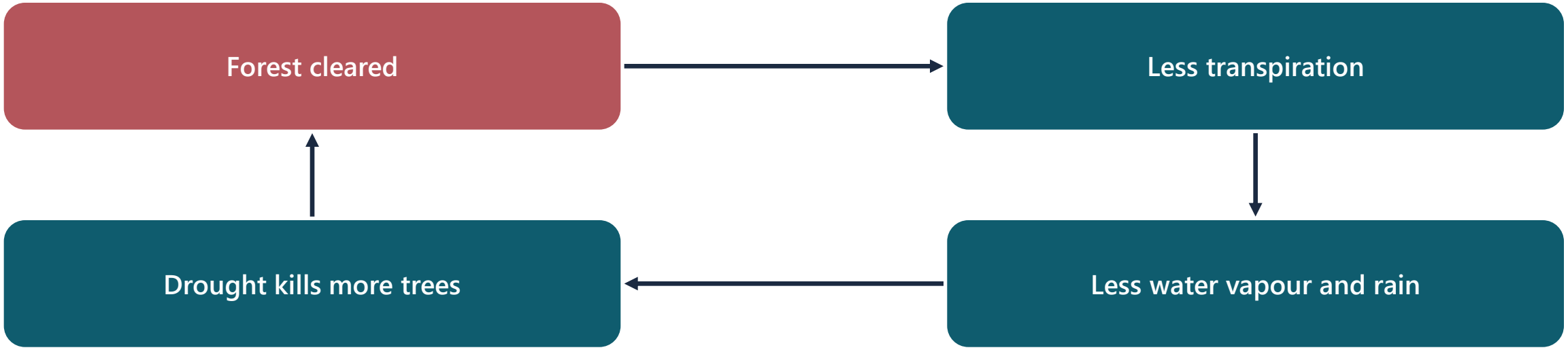
**Activity**

Timber harvesting, poaching a top predator, soil erosion: which requirement does each put at risk?

# Tipping points: the Amazon

D4.2.3

A **tipping point** is where a system shifts to a new state and may not recover. **Positive feedback** drives it.

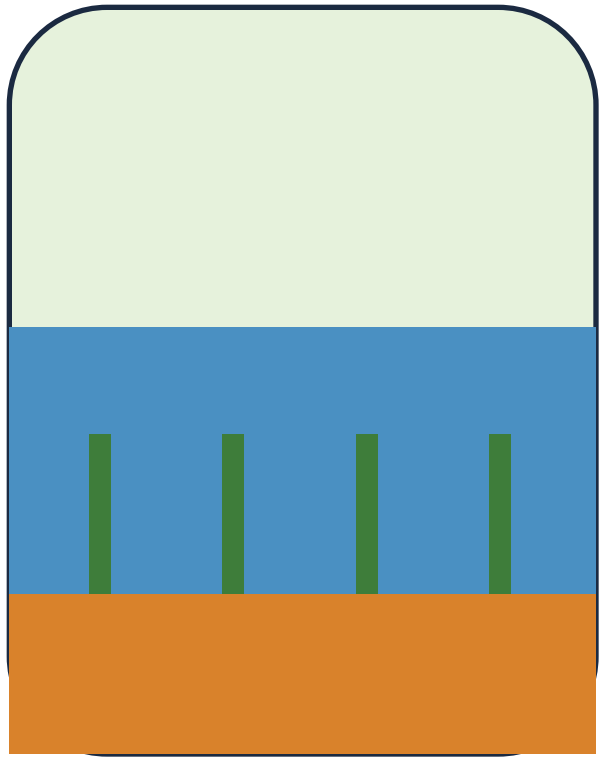



**Big question**

Large rainforests make much of their own rain. Clear enough and parts may shift to savanna; scientists disagree on when.

# Modelling stability: mesocosms





sealed jar: light in, heat out

A **mesocosm** is a small enclosed model ecosystem. Sealed vessels stop matter crossing but let energy through, so you can test what keeps it stable.

### Design and ethics

Clear or opaque? Which trophic levels? Should animals be included if they might suffer?

# Practice: stability

D4.2.1–3

**List**

Four requirements for stability.

**Explain**

Positive feedback in the Amazon.

**Design**

A mesocosm to test light and stability.

**Use**

stability, tipping point, positive feedback, mesocosm

# Check-in: class 1

D4.2.1–3

1. Define stability.
2. List four requirements for a stable ecosystem.
3. Define tipping point.
4. Explain how deforestation can feed back on itself.
5. Describe a mesocosm.

**How to use this**

Answer without your notes, then check against slides 3–8.

# Keystone species

D4.2.4



Beaver dam. Sean Dudley, public domain.

A **keystone species** has an effect on community structure far larger than its numbers suggest. Remove it and the ecosystem may collapse.

Beavers build dams that create wetlands for many species.

### Swiss example

Amphipods in rivers break down leaf litter and feed fish.

# Wolves in Yellowstone

D4.2.4



Wolf. NPS, public domain.

### Debate

Scientists still argue how much of the recovery is due to wolves and how much to drought, hunting and bears.

Wolves return (1995) → Elk fall; behaviour changes → Willow and aspen recover → Beavers and birds return

### Key idea

Removing or restoring a keystone predator changes the whole community.

# Is harvesting sustainable?

D4.2.5

## The rule

Harvesting is sustainable if its rate is lower than the rate of replacement.




## Ask

How fast does it reproduce? How fast is it removed?

# Timber and cod compared

D4.2.5

| | Timber in Finland | Atlantic cod |
|---|---|---|
| **Replacement** | Regrowth required by law | Depends on adults left to breed |
| **Assessment** | Growth and felling rates | Surveys and catch records |
| **Outcome** | Forest maintained | Stocks collapsed (Canada, 1990s) |



Atlantic cod. Public domain.

### Why cod collapsed

Catches exceeded breeding; large breeding adults were removed first. Stocks have still not fully recovered.

# Sustainability of agriculture

D4.2.6

| Factor | Issue |
|---|---|
| **Soil erosion** | Ploughed soil blown and washed away |
| **Leaching** | Nutrients washed into rivers |
| **Fertilizers** | Limited supply; energy to make |
| **Pesticides** | Harm other organisms |
| **Carbon footprint** | Machines, transport, glasshouses |

**Discuss**

Which changes would make a farm more sustainable, and at what cost?

# Working with data: a fish stock

D4.2.5

Breeding stock and catch of a fish (thousand tonnes). Invented values; realistic pattern of collapse.

| Year | Breeding stock | Catch |
|---|---|---|
| **1970** | 1200 | 250 |
| **1980** | 900 | 300 |
| **1990** | 400 | 280 |
| **2000** | 80 | 10 |

1. Describe the trend in stock.
2. When did harvesting become unsustainable? Explain.
3. Why did catch fall to 10?

# Check-in: class 2

D4.2.4–6

1. Define keystone species.
2. Give an example and its effect.
3. State the rule for sustainable harvesting.
4. Explain why cod stocks collapsed.
5. Give three ways farming can be unsustainable.

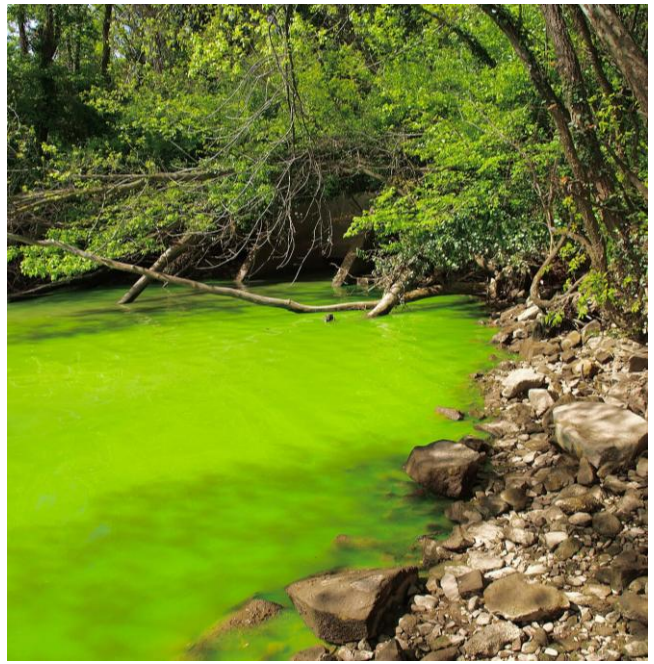
**How to use this**

Answer without your notes, then check against slides 10–15.

# Eutrophication

D4.2.7

Fertilizer leaches in → Algal bloom blocks light → Plants below die → **Bacteria decompose; use $O_2$** → Fish suffocate



Algal bloom in the Potomac. A. Trubetskoy, CC BY-SA 3.0.

## BOD

**Biochemical oxygen demand** is the oxygen microorganisms use to break down organic matter. High BOD means heavy organic pollution and too little oxygen for animals.

# Working with data: oxygen downstream

D4.2.7

Dissolved oxygen and BOD downstream of a farm drain. Invented values; realistic pattern.

| Distance / km | $O_2$ / mg $dm^{-3}$ | BOD / mg $dm^{-3}$ |
|---|---|---|
| **Upstream** | 9.5 | 2 |
| **0.5** | 4.0 | 12 |
| **2** | 6.0 | 7 |
| **5** | 8.8 | 3 |

1. Describe both trends.
2. Explain the low $O_2$ at 0.5 km.
3. Explain the recovery downstream.

# Biomagnification

D4.2.8




**Why**

Fat-soluble, not excreted; each consumer eats many prey.

# Accumulate and magnify

D4.2.8

| | Bioaccumulation | Biomagnification |
|---|---|---|
| **Meaning** | Builds up in one organism over its life | Rises at each trophic level |
| **Why** | Cannot be excreted (fat-soluble) | Each consumer eats many prey |

### Examples

**DDT** thinned eggshells of birds of prey. **Mercury** builds up in large predatory fish such as sharks and swordfish.

# Plastics in the oceans

D4.2.9



Albatross chick with plastic. Duncan Wright, public domain.



Microplastics on a beach. Ifremer, CC BY 4.0.

| | Macroplastics | Microplastics |
|---|---|---|
| **Size** | > 5 mm | < 5 mm |
| **Harm** | Entangle; fill stomachs | Eaten by small animals; pass along chains |

### Why they persist

Most plastics do not biodegrade. Albatrosses feed plastic to chicks; turtles mistake bags for jellyfish.

# Rewilding

D4.2.10

| Restore | Reconnect | Remove |
| --- | --- | --- |
| Apex predators and keystone species | Habitats across large areas | Livestock and invasive species |

**Examples**

Hinewai Reserve (New Zealand); Rewilding Europe in the Iberian Highlands and Danube Delta; Swiss National Park.

**Discuss**

Farmers may oppose returning wolves. How should conservation and livelihoods be balanced?

# Practice: pollution and recovery

D4.2.7–10

**Explain**

Why bacteria, not algae, use up the oxygen.

**Calculate**

DDT magnification from plankton to bird.

**Evaluate**

Rewilding with large predators.

**Use**

eutrophication, BOD, biomagnification, rewilding

# Working with data: mercury in a food chain

D4.2.8

Mercury in a lake food chain (ppb). Rounded values typical of published studies.

| Organism | Mercury / ppb |
|---|---|
| **Water** | 0.01 |
| **Algae** | 10 |
| **Small fish** | 60 |
| **Large fish** | 400 |
| **Fish-eating bird** | 2 000 |

1. How many times higher in the bird than the algae?
2. Explain the increase.
3. Suggest advice for people eating fish here.

# Check-in: class 3

D4.2.7–10

1. Explain how eutrophication kills fish.
2. Define BOD.
3. Distinguish bioaccumulation and biomagnification.
4. Distinguish macroplastics and microplastics.
5. Outline three rewilding methods.

**How to use this**

Answer without your notes, then check against slides 17–24.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **Eutrophication poisons fish.** | They die from lack of oxygen. |
| **Algae use up the oxygen.** | Bacteria decomposing dead algae do. |
| **Keystone species are the most common.** | Their effect is large for their numbers. |
| **Stable means unchanging.** | Populations fluctuate within limits. |

# Summary: stability and change

D4.2.1–10

| | Key idea |
|---|---|
| **D4.2.1–2** | Stability and its requirements; mesocosms |
| **D4.2.3** | Tipping points: Amazon feedback |
| **D4.2.4** | Keystone species |
| **D4.2.5–6** | Sustainable harvesting and farming |
| **D4.2.7–8** | Eutrophication, BOD, biomagnification |
| **D4.2.9–10** | Plastics; rewilding |

# Check that you can...

1. **Outline** stability and **explain** tipping points
2. **Explain** keystone species
3. **Evaluate** sustainability of harvesting and farming
4. **Explain** eutrophication and biomagnification
5. **Outline** plastic effects and rewilding

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. State the condition that makes harvesting of a species sustainable.
2. Outline how leaching of fertilizer can lead to the death of fish.
3. Distinguish between bioaccumulation and biomagnification.

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